

Honors Precalculus — Unit 1: Parent Functions

Standards-Based Grading Rubric

These are the skills students should master by the end of the unit — objectives are not tied to a single class period; several may stretch across multiple days or be woven together.

A blank cell means that skill is only being assessed up through the level(s) shown — for example, a blank Advanced column means students are not assessed beyond Intermediate for that objective. |
TEKS source: TEKS RS Precalculus, Strand P.2 (Functions), tcmpc.org standards view.

Obj #	Learning Target	TEKS	Basic	Intermediate	Advanced
1-1	Identify parent functions	P.2.F	Identify the parent function family (match equation to graph) for the Algebra II functions: exponential, logarithmic, rational, polynomial, and power.	Identify and sketch, from its equation, every parent function required in Precalculus, including trigonometric, inverse trigonometric, and piecewise-defined functions (including step functions).	
1-2	Determine and analyze key features	P.2.I	Identify at least 5 of the 7 key features (domain, range, symmetry, relative maximum, relative minimum, zeros, asymptotes) from a graph.	Identify all 7 key features from a graph, table, or equation for exponential, logarithmic, rational, polynomial, power, trigonometric, inverse trigonometric, and piecewise-defined functions.	Identify all 7 key features and express domain, range, and other features using proper interval, set, and inequality notation; justify zeros and asymptotes algebraically from the equation.
1-3	Identify increasing / decreasing intervals	P.2.I	Given a graph, state where the function is increasing or decreasing using informal, verbal language (e.g., “goes up,” “goes down”).	Given a graph or equation, determine and state the intervals where the function is increasing or decreasing.	State increasing/decreasing intervals using proper interval notation (e.g., $(-1, 3]$) and set notation (e.g., $\{x \mid x \in \mathbb{R}, -1 < x \leq 3\}$) directly from the equation, without graphing technology.
1-4	Describe end behavior	P.2.J	Using a table or graph, describe the end behavior of a function in words (e.g., “as x gets larger, y gets larger”).	Describe the end behavior of exponential, logarithmic, rational, polynomial, and power functions from a graph or table using infinity notation ($x \rightarrow \infty$, $f(x) \rightarrow \infty$).	
1-5	Classify even and odd functions	P.2.D	Visually identify the symmetry of a graph (y -axis, origin, or neither).	Classify a function as even, odd, or neither from its graph, and explain the classification using the definitions $f(-x) = f(x)$ (even) or $f(-x) = -f(x)$ (odd).	Algebraically test $f(-x)$ to prove whether a function given in equation form is even, odd, or neither, and connect the algebraic result to the graph’s symmetry.
1-6	Analyze discontinuities	P.2.L, P.2.M	Determine whether a graph is continuous or discontinuous and identify the x -value(s) where a discontinuity occurs.	Classify each discontinuity as removable (hole), jump, or infinite (vertical asymptote) from a graph or table.	Describe the left-sided and right-sided behavior of a function around a discontinuity using proper notation ($x \rightarrow c^-$, $x \rightarrow c^+$) and determine the type of discontinuity algebraically from a rational or piecewise equation.
1-7	Graph function transformations	P.2.G	Identify 2 of the 4 transformation parameters (a , b , c , or d) applied to a parent function — $af(x)$, $f(x) + d$, $f(x - c)$, $f(bx)$ — from its equation or graph.	Identify all 4 transformation parameters and describe their effect on the graph (vertical/horizontal stretch, compression, reflection, or shift).	Identify all 4 transformation parameters using proper function notation and graph combined transformations of a parent function without technology.